

Salt Analysis Inference Tables

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User prompt: into this pdf for each ion add an "inference" tab too.

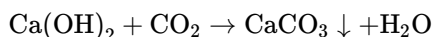
Response:

Part I: Acidic Radicals (Anions)

A. Dilute H_2SO_4 Group

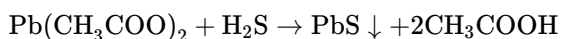
1. Carbonate (CO_3^{2-})

- **Preliminary Test:** Add a few drops of dilute H_2SO_4 to a small quantity of the given salt in a test tube.
 - **Observation:** Brisk effervescence occurs with the evolution of a colorless, odorless gas.
 - **Inference:** Carbonate (CO_3^{2-}) ion is indicated.
- **Confirmatory Test (Lime Water Test):** Pass the evolved gas through freshly prepared lime water.
 - **Observation:** The lime water turns milky due to the formation of insoluble calcium carbonate.
 - **Inference:** Carbonate (CO_3^{2-}) ion is confirmed.
 - **Reaction:**



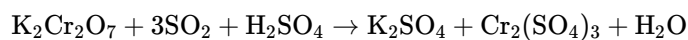
2. Sulphide (S^{2-})

- **Preliminary Test:** Add dilute H_2SO_4 to the salt and warm slightly.
 - **Observation:** A colorless gas with a distinct rotten egg smell is evolved.
 - **Inference:** Sulphide (S^{2-}) ion is indicated.
- **Confirmatory Test (Lead Acetate Test):** Bring a piece of filter paper moistened with lead acetate solution to the mouth of the test tube.
 - **Observation:** The filter paper turns shining black.
 - **Inference:** Sulphide (S^{2-}) ion is confirmed.
 - **Reaction:**



3. Sulphite (SO_3^{2-})

- **Preliminary Test:** Add dilute H_2SO_4 to the salt.
 - **Observation:** A colorless gas with a suffocating smell of burning sulphur is evolved.
 - **Inference:** Sulphite (SO_3^{2-}) ion is indicated.
- **Confirmatory Test (Potassium Dichromate Test):** Bring a filter paper dipped in acidified potassium dichromate ($\text{K}_2\text{Cr}_2\text{O}_7$) solution near the mouth of the test tube.
 - **Observation:** The filter paper turns from orange to green.
 - **Inference:** Sulphite (SO_3^{2-}) ion is confirmed.
 - **Reaction:**

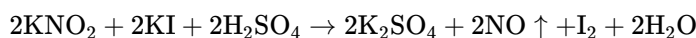


4. Nitrite (NO_2^-)

- **Preliminary Test:** Add dilute H_2SO_4 to the salt.
 - **Observation:** A reddish-brown gas with a pungent smell is evolved.
 - **Inference:** Nitrite (NO_2^-) ion is indicated.

- **Confirmatory Test (Starch Iodide Test):** Prepare an aqueous solution of the salt. Add a few drops of dilute H_2SO_4 , followed by potassium iodide (KI) solution and freshly prepared starch solution.

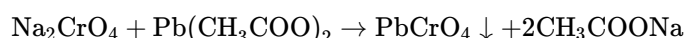
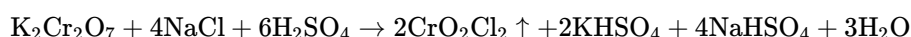
- **Observation:** A deep blue-black coloration appears.
- **Inference:** Nitrite (NO_2^-) ion is confirmed.
- **Reaction:**



B. Concentrated H_2SO_4 Group

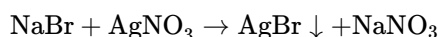
1. Chloride (Cl^-)

- **Preliminary Test:** Add concentrated H_2SO_4 to the salt and heat.
 - **Observation:** A colorless, pungent-smelling gas evolves which gives dense white fumes when a glass rod dipped in NH_4OH is brought near it.
 - **Inference:** Chloride (Cl^-) ion is indicated.
- **Confirmatory Test (Chromyl Chloride Test):** Mix the salt with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and add concentrated H_2SO_4 . Heat the mixture and pass the red vapors evolved into NaOH solution. Add acetic acid and lead acetate solution to the resulting yellow solution.
 - **Observation:** A yellow precipitate of lead chromate forms.
 - **Inference:** Chloride (Cl^-) ion is confirmed.
 - **Reactions:**



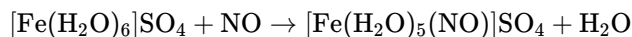
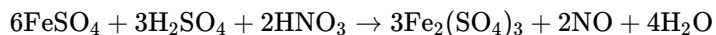
2. Bromide (Br^-)

- **Preliminary Test:** Add concentrated H_2SO_4 to the salt and heat.
 - **Observation:** Reddish-brown vapors of bromine gas are evolved.
 - **Inference:** Bromide (Br^-) ion is indicated.
- **Confirmatory Test (Silver Nitrate Test):** To the aqueous salt extract, add dilute HNO_3 followed by silver nitrate (AgNO_3) solution.
 - **Observation:** A pale yellow precipitate forms which is partially soluble in ammonium hydroxide (NH_4OH) solution.
 - **Inference:** Bromide (Br^-) ion is confirmed.
 - **Reaction:**



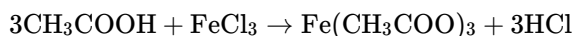
3. Nitrate (NO_3^-)

- **Preliminary Test:** Add concentrated H_2SO_4 to the salt and heat. Add a few copper turnings.
 - **Observation:** The evolution of dense reddish-brown fumes (NO_2) intensifies.
 - **Inference:** Nitrate (NO_3^-) ion is indicated.
- **Confirmatory Test (Brown Ring Test):** Take the aqueous solution of the salt in a test tube. Add freshly prepared ferrous sulphate (FeSO_4) solution to it in equal volume. Carefully pour concentrated H_2SO_4 down the inner side of the inclined test tube.
 - **Observation:** A dark brown ring forms at the junction of the two liquids.
 - **Inference:** Nitrate (NO_3^-) ion is confirmed.
 - **Reactions:**



4. Acetate (CH_3COO^-)

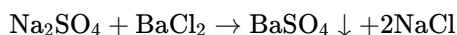
- **Preliminary Test:** Take a small amount of salt on a watch glass, add a few drops of concentrated H_2SO_4 and rub it with a glass rod.
 - **Observation:** A characteristic smell of vinegar is observed.
 - **Inference:** Acetate (CH_3COO^-) ion is indicated.
- **Confirmatory Test (Ferric Chloride Test):** Add neutral ferric chloride (FeCl_3) solution to the aqueous salt extract.
 - **Observation:** A deep red or blood-red coloration is produced.
 - **Inference:** Acetate (CH_3COO^-) ion is confirmed.
 - **Reaction:**



C. Independent Group

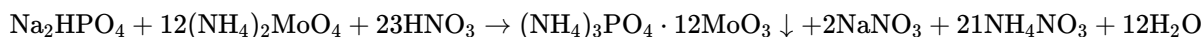
1. Sulphate (SO_4^{2-})

- **Test:** To the aqueous solution of the salt, add dilute HCl and boil. Then, add barium chloride (BaCl_2) solution.
 - **Observation:** A thick white precipitate forms which remains insoluble even upon adding concentrated HCl .
 - **Inference:** Sulphate (SO_4^{2-}) ion is confirmed.
 - **Reaction:**



2. Phosphate (PO_4^{3-})

- **Test:** To the aqueous salt solution, add concentrated HNO_3 and boil. Add ammonium molybdate solution to the hot mixture and heat again.
 - **Observation:** A canary yellow precipitate of ammonium phosphomolybdate is formed.
 - **Inference:** Phosphate (PO_4^{3-}) ion is confirmed.
 - **Reaction:**

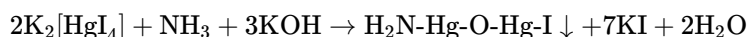


Part II: Basic Radicals (Cations)

Group 0

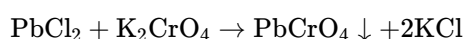
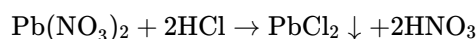
1. Ammonium (NH_4^+)

- **Preliminary Test:** Heat a small quantity of the original salt with sodium hydroxide (NaOH) solution.
 - **Observation:** A colorless gas with a pungent, choking smell of ammonia is evolved.
 - **Inference:** Ammonium (NH_4^+) ion is indicated.
- **Confirmatory Test (Nessler's Reagent Test):** Pass the evolved gas through Nessler's reagent (or add the reagent directly to the salt solution).
 - **Observation:** A brown precipitate or deep brown coloration is obtained.
 - **Inference:** Ammonium (NH_4^+) ion is confirmed.
 - **Reactions:**

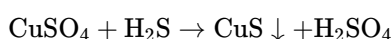


Group I1. Lead (Pb^{2+})

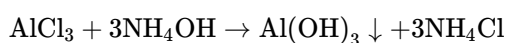
- **Preliminary Test:** Prepare the Original Solution (O.S.) of the salt in water. Add dilute HCl .
 - **Observation:** A white precipitate of lead chloride forms, which dissolves on heating and reappears on cooling.
 - **Inference:** Lead (Pb^{2+}) ion (Group I) is indicated.
- **Confirmatory Test (Potassium Chromate Test):** Dissolve the white precipitate in hot water. Add potassium chromate (K_2CrO_4) solution.
 - **Observation:** A brilliant yellow precipitate of lead chromate forms.
 - **Inference:** Lead (Pb^{2+}) ion is confirmed.
 - **Reactions:**

**Group II**1. Copper (Cu^{2+})

- **Preliminary Test:** Pass H_2S gas through the acidified original solution.
 - **Observation:** A black precipitate of CuS forms. Dissolve this precipitate in 50% nitric acid and boil off the H_2S gas.
 - **Inference:** Copper (Cu^{2+}) ion (Group II) is indicated.
- **Confirmatory Test (Ammonia Test):** To the above clear solution, add ammonium hydroxide (NH_4OH) solution dropwise, then in excess.
 - **Observation:** A pale blue precipitate forms initially, which dissolves in excess reagent to yield a deep blue (ink blue) colored solution.
 - **Inference:** Copper (Cu^{2+}) ion is confirmed.
 - **Reaction:**

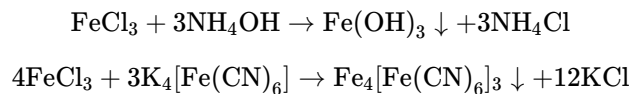
**Group III**1. Aluminium (Al^{3+})

- **Preliminary Test:** Boil off H_2S from the Group II filtrate. Add a few drops of concentrated HNO_3 and boil. Cool, add solid NH_4Cl , and then add NH_4OH in excess.
 - **Observation:** A gelatinous white precipitate forms. Dissolve it in dilute HCl .
 - **Inference:** Aluminium (Al^{3+}) ion (Group III) is indicated.
- **Confirmatory Test (Lake Test):** To the dissolved solution, add 2 drops of blue litmus solution, followed by NH_4OH dropwise until the solution is alkaline.
 - **Observation:** A blue mass (precipitate) floats suspended in a colorless solution.
 - **Inference:** Aluminium (Al^{3+}) ion is confirmed.
 - **Reaction:**

2. Ferric (Fe^{3+})

- **Preliminary Test:** Follow the same precipitation steps as for Aluminium.
 - **Observation:** A reddish-brown precipitate forms. Dissolve it in dilute HCl .
 - **Inference:** Ferric (Fe^{3+}) ion (Group III) is indicated.

- **Confirmatory Test (Prussian Blue Test):** To the dissolved solution, add potassium ferrocyanide ($K_4[Fe(CN)_6]$) solution.
 - **Observation:** A deep blue (Prussian blue) precipitate or coloration is formed.
 - **Inference:** Ferric (Fe^{3+}) ion is confirmed.
 - **Reactions:**



Group IV

1. Zinc (Zn^{2+})

- **Preliminary Test:** Pass H_2S gas through the ammoniacal solution of Group III.
 - **Observation:** A dirty white or greyish-white precipitate forms. Dissolve this in dilute HCl and boil off H_2S .
 - **Inference:** Zinc (Zn^{2+}) ion (Group IV) is indicated.
- **Confirmatory Test (NaOH Test):** Add sodium hydroxide ($NaOH$) solution dropwise, and then in excess, to the clear solution.
 - **Observation:** A white precipitate forms initially, which dissolves completely in excess $NaOH$ to form a clear soluble solution of sodium zincate.
 - **Inference:** Zinc (Zn^{2+}) ion is confirmed.
 - **Reactions:**

